

ENTERPRISE VULNERABILITY ASSESSMENT USING OPENVAS

A Practical Deployment and Operations Reference

Vulnerability Management • Asset Discovery • Risk Prioritization

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Environment details, hostnames, and IP addressing throughout this document are illustrative and have been generalized for public distribution.

1. Introduction

OpenVAS (Open Vulnerability Assessment Scanner) is the scanning engine at the core of the Greenbone Vulnerability Management platform. It performs authenticated and unauthenticated network scans against target hosts, evaluating each one against a continuously updated feed of Network Vulnerability Tests (NVTs) covering missing patches, misconfigurations, weak protocols, default credentials, and known CVEs. Findings are scored using the Common Vulnerability Scoring System (CVSS) and organized into reports that security teams can review, filter, prioritize, and export.

This document describes a practical enterprise deployment of OpenVAS: how targets are organized, how scans are configured and executed, how results are interpreted and prioritized, and the operational procedures that keep a recurring vulnerability management program running reliably. The intent is to demonstrate a complete, end-to-end vulnerability management workflow suitable as a reference for teams standing up or operating an OpenVAS deployment.

1.1 Purpose and Scope

The scope of this guide covers the full lifecycle of a vulnerability assessment: environment preparation, asset and target definition, scan configuration selection, task execution, monitoring, result interpretation, risk-based prioritization, and reporting. It also covers common operational issues and the best practices that reduce noise and increase the reliability of a scanning program over time.

1.2 Intended Audience

This guide is written for security analysts, system administrators, and engineers responsible for operating a vulnerability scanning program, as well as for readers evaluating OpenVAS as a candidate platform for their own environment.

1.3 Complementary Tooling

A vulnerability scanner answers a point-in-time question: what weaknesses exist on a host right now. It is commonly deployed alongside continuous, host-based monitoring platforms (such as SIEM or EDR/XDR tooling) that answer a different question: what is happening on a host right now. The two are complementary rather than interchangeable. Findings from a scan can inform which continuous-monitoring rules or alerts are prioritized for a given host, and an anomalous alert from continuous monitoring can, in turn, trigger an out-of-cycle scan to confirm whether a newly discovered vulnerability was the entry point.

2. OpenVAS Architecture

A Greenbone/OpenVAS deployment is composed of a small number of cooperating services. Understanding how they relate clarifies why scans are configured the way they are and where to look when something needs troubleshooting.

2.1 High-Level Data Flow

- An analyst defines Targets (the hosts to be scanned) and Scan Configurations (which tests to run) through the web interface.
- A Task binds a Target, a Scanner, and a Scan Configuration into a runnable, repeatable job.
- When a Task is started, the Manager hands the job to a Scanner, which executes the relevant NVTs from the current feed against the target hosts.
- Findings stream back to the Manager as the scan progresses and are compiled into a Report once the scan completes.
- Reports are reviewed through the interface or exported in PDF, CSV, or HTML format for distribution or further processing.

2.2 Deployment Topology

In a typical enterprise deployment, the scanner is positioned with network visibility into every segment it is expected to assess. Where segments are isolated by firewall or VLAN boundaries, explicit rules must permit the scanner's probe traffic to reach target hosts and permit responses back, or host discovery and scan coverage will silently degrade even though the task itself reports as completed.

3. Components

3.1 Greenbone Security Assistant (GSA)

The web-based user interface used to configure targets, scan configurations, tasks, and to review reports. All operator interaction described in this guide takes place through the GSA.

3.2 Scanner

The engine that actively executes NVTs against target hosts. A standard deployment includes the OpenVAS Scanner (full active scanning) and, optionally, a CVE Scanner for fast, passive-style CVE matching without full active probing.

3.3 Feed

The continuously updated collection of NVTs, CVE data, and CERT advisories that scans are evaluated against. Feed currency directly determines whether a scan can detect recently disclosed vulnerabilities; a stale feed undermines the value of every scan run against it.

3.4 Manager

The service responsible for coordinating scanners, storing configuration objects (targets, tasks, configs, credentials), and compiling scan results into reports.

3.5 Targets

Named groups of hosts, defined by IP address, range, or CIDR block, together with an associated port list and, optionally, authentication credentials for authenticated scanning.

3.6 Tasks

The object that ties a Target, a Scanner, and a Scan Configuration together into something that can actually be run, scheduled, and repeated.

3.7 Reports

The output of a completed scan task: every finding discovered, organized by severity, host, port, and detection confidence, with drill-down detail available per finding.

3.8 NVT Feed

The specific subset of the Feed containing the executable test definitions (Network Vulnerability Tests) themselves — the checks a Scan Configuration selects from when a scan runs.

4. Deployment Overview

Rather than scanning an entire network as a single undifferentiated block, this deployment organizes targets by logical segment — reflecting how the environment is structured operationally. This keeps scan results attributable to a specific owner, speeds up remediation, and prevents one segment's high-volume, low-severity noise from burying another segment's genuinely critical findings.

4.1 Why Assets Are Grouped by Segment

- **Accountability** — when a finding is reported, it is immediately clear which segment owns the affected host and who should action remediation.
- **Scan tuning** — different segments run different technology stacks (web servers behave very differently from embedded IoT devices), so each group can eventually be scanned with a profile and schedule suited to its own risk profile.
- **Reporting clarity** — periodic reports can be filtered and distributed per segment rather than forcing every stakeholder to search an organization-wide report for their section.

4.2 Reference Segment Layout

The table below illustrates a representative segment layout used throughout the remainder of this guide. Addressing is illustrative.

Segment Name	Illustrative IP Range	Purpose
Production Server	10.10.0.10	Primary infrastructure / core server host
Engineering Segment	10.10.1.10 – 10.10.1.59	Engineering workstations and lab hosts

Segment Name	Illustrative IP Range	Purpose
Development Segment	10.10.1.60 – 10.10.1.64	Application development workstations and dev servers
Design Segment	10.10.1.65 – 10.10.1.69	Design team workstations
Marketing Segment	10.10.1.70 – 10.10.1.74	Marketing workstations
IoT / Operations Segment	10.10.1.75 – 10.10.1.79	IoT devices and operations-lead endpoints
Unassigned Segment	10.10.1.80 – 10.10.1.108	Temporary and miscellaneous unassigned hosts
Administrative Segment	10.10.2.5 – 10.10.2.10	Administrative and management workstations
Surveillance Segment	10.10.2.41 – 10.10.2.50	CCTV / surveillance device network

Note: Segment tables should be kept current. When infrastructure changes, update both the live Targets configuration and this reference table together so they never drift apart.

5. Environment Preparation

Before scanning begins, three prerequisites should be confirmed: scanner reachability, feed currency, and network access.

5.1 Scanner Readiness

Confirm the scanner shows as reachable under Configuration → Scanners before relying on it for any task. A standard deployment exposes both an OpenVAS Scanner and a CVE Scanner as distinct, independently selectable engines.

5.2 Feed Synchronization

Confirm the NVT feed has synced recently under Administration → Feed Status. Scanning against a stale feed produces results that miss recently disclosed vulnerabilities, undermining the value of the entire exercise.

5.3 Network Access

Where target segments sit behind firewall or routing boundaries, confirm the scanner's address is explicitly permitted to reach the target range, and that responses are permitted back. Missing rules typically manifest as zero hosts discovered rather than an explicit error.

6. Target Configuration

Targets must exist before a scan task can be created. This section covers creating, editing, and validating a target.

6.1 Creating a New Target

1. Navigate to Configuration → Targets in the left-hand menu.
2. Click the new-object icon near the top-left of the Targets list.
3. Enter a clear, segment-aligned Name — this name will appear throughout tasks, reports, and dashboards, so keep naming consistent across the environment.
4. Under Hosts, choose how IP addresses will be entered: manually (comma-separated addresses or ranges) or from a hosts file.
5. Enter the address range. A single host can be entered as a plain address (for example, 10.10.0.10); a range can be entered as a hyphenated range (for example, 10.10.1.60-10.10.1.64) or CIDR notation.
6. Select a Port List — a comprehensive default such as “All IANA assigned TCP” is the standard choice for general scanning.
7. Optionally attach Credentials to enable authenticated checks.
8. Click Create.

6.2 Reference Examples

Example Target	Example Range	Host Count
Design-Segment	10.10.1.65 – 10.10.1.69	5
Development-Segment	10.10.1.60 – 10.10.1.64	5
Engineering-Segment	10.10.1.10 – 10.10.1.59	50

6.3 Editing Targets

From the Targets list, use the edit icon in the Actions column to modify the Hosts field or assigned Port List, then save.

Caution: Editing a target referenced by an existing task changes what that task scans on its next run. It does not retroactively change historical reports. Always confirm which tasks reference a target before editing it broadly.

6.4 Validating Targets

- Open the target detail page and confirm the IP count matches the number of addresses expected for that range.
- Cross-check the range against the segment's actual current asset list — ranges drift over time as hardware is added, retired, or re-addressed.
- Run a quick Host Discovery scan against the target as a smoke test; it should report live hosts matching expectations within minutes.

7. Port Lists

A port list defines which TCP/UDP ports a scan will probe. Port lists can be broad (all IANA-assigned ports) or narrow (a handful of specific ports), depending on scan goals.

7.1 Choosing Breadth

Most general vulnerability scans use a comprehensive default port list so nothing is missed; narrower, custom port lists are reserved for quick, targeted checks where scan duration matters more than exhaustive coverage.

7.2 What to Review

Review the total port count and the TCP/UDP split to confirm the right breadth of coverage is selected for the scan profile in use. A port list with only a handful of entries attached to a task intended as a comprehensive sweep is a common, easily overlooked cause of incomplete results.

8. Scan Configurations

Scan Configurations define which NVT families are included in a scan and how thoroughly they run — the difference between a five-minute discovery sweep and a multi-hour deep assessment.

Discovery

Purpose: A broad reconnaissance configuration that identifies live hosts, open ports, and running services without performing deep vulnerability checks.

When to use: Early in onboarding a new target, or whenever the goal is to build an accurate asset/service inventory before running a full scan.

Advantages: Fast; covers a very large number of NVT families for service identification; low impact on target systems.

Limitations: Does not produce meaningful CVSS-scored findings — most results appear as Log severity or N/A, which is expected and not an error.

Host Discovery

Purpose: A minimal configuration focused purely on determining which addresses in a range currently respond.

When to use: As a quick smoke test after creating or editing a target, or to validate a range before committing to a longer scan.

Advantages: Extremely fast (often seconds to a couple of minutes); very low network impact.

Limitations: Provides no vulnerability information whatsoever — it is purely a liveness check.

System Discovery

Purpose: An expanded discovery configuration that, beyond host liveness, attempts to fingerprint operating systems and key system-level services.

When to use: When building or refreshing an operating-system inventory across a target group.

Advantages: Populates OS and CPE data more reliably than basic Host Discovery; still significantly faster than a full scan.

Limitations: Still not a substitute for a real vulnerability scan — findings remain mostly informational.

Full and Fast

Purpose: The standard, comprehensive vulnerability scan configuration. It runs the largest set of NVT families and tests, using previously collected information to optimize test selection.

When to use: The default profile for routine scanning against every segment target — the primary, recommended profile for ongoing operational use.

Advantages: Broadest coverage of any built-in profile; produces the CVSS-scored, actionable findings that reports and remediation tracking depend on.

Limitations: Takes meaningfully longer than any discovery-only profile; generates a larger volume of results to triage, including many informational Log-level findings alongside true vulnerabilities.

Log4Shell

Purpose: A narrow, purpose-built configuration containing checks specifically for Log4j-related vulnerabilities, most notably CVE-2021-44228.

When to use: For rapid, targeted sweeps across the environment when a specific high-profile vulnerability needs an immediate answer, without waiting for a full scan cycle.

Advantages: Very fast compared to Full and Fast since it runs a small, focused test set; ideal for emergency out-of-cycle checks.

Limitations: Only checks for the specific vulnerability family it targets — a clean result says nothing about exposure to any other vulnerability class.

9. Creating Scan Tasks

Once a target exists, a task can be created to run scans against it. Full and Fast is used here as the primary worked example, since it is the standard profile for comprehensive periodic scanning.

9.1 Creating a Task

9. Navigate to Scans → Tasks.

10. Click the new-object icon to open the New Task dialog.

11. Enter a Name that mirrors the target it scans (for example, “Design-Segment-Scan”) so the task list stays easy to scan visually.
12. Optionally add a Comment describing the scan's purpose.

9.2 Selecting a Target and Scanner

13. In the Scan Targets dropdown, select the previously created target.
14. Confirm the host count shown next to the target matches expectations before proceeding.
15. In the Scanner dropdown, select the OpenVAS Default scanner for a standard full active scan, or the CVE Scanner when only a fast, database-only check is needed.

9.3 Choosing a Scan Configuration

16. In the Scan Config dropdown, select Full and Fast for a standard comprehensive scan, or a narrower profile (Discovery, Host Discovery, System Discovery, Log4Shell) as appropriate — see Section 8.

9.4 Saving the Task

17. Review all selections: Name, Target, Scanner, and Scan Config.
18. Click Create to save the task; it appears in the Tasks list with status New.

10. Running Vulnerability Assessments

With a task created, the assessment is launched from the Tasks list.

19. Click the Start (play) icon in the Actions column to launch the scan immediately, or configure a Schedule under Configuration → Schedules to run it automatically on a recurring basis.
20. The task status changes to Requested, then Running, while the scan executes.

Note: A successfully launched task displays a confirmation banner such as “Task [name] scan started successfully” at the top of the screen.

11. Monitoring Scan Progress

Every task moves through a defined set of statuses from creation to completion. Knowing what each status means makes it possible to distinguish a healthy in-progress scan from one that needs attention.

Status	Meaning	Action Needed
New	Task has been created but never started.	None — start when ready, or confirm this is intentional.
Requested	Start has been triggered; the task is queued, waiting for the scanner to begin.	Wait. If stuck for an extended period, check scanner availability (Section 19).
Running	The scan is actively executing against the target's	None — allow it to complete. Duration

Status	Meaning	Action Needed
	hosts.	depends on host count and profile.
Done	The scan completed successfully and a report is available.	Open the report and begin the review workflow (Section 12).
Stopped	The scan was manually stopped before completion.	Restart if a full result set is needed; otherwise review the partial report.
Interrupted	The scan stopped unexpectedly, typically due to a scanner or connectivity issue.	Investigate the cause (Section 19), then re-run the task.

11.1 Verifying Successful Completion

- Confirm the task status reads Done, not Stopped or Interrupted.
- Confirm the Reports count incremented by one and the Last Report timestamp is current.
- Open the new report and check the Results tab count is non-zero and consistent with the target's host count.
- Check that the report's overall Severity badge is populated with a real value rather than N/A where vulnerabilities would normally be expected.

12. Viewing Reports

Once a scan is Done, findings can be reviewed from several angles depending on the question being asked.

12.1 Reports Page

Scans → Reports lists every completed report with severity totals broken out by column (Critical, High, Medium, Low, Log, False Positive). This is the right starting point when the question is which scans need attention today — scan down the Critical and High columns across all rows rather than opening every report individually.

12.2 Results Page

Scans → Results flattens every finding from every report into one searchable list. This is the right tool when the question concerns a specific finding type across the whole environment, rather than one specific segment's report.

12.3 Vulnerabilities Page

Scans → Vulnerabilities groups findings by vulnerability type and shows host counts. Use this when prioritizing remediation by blast radius — fixing a vulnerability affecting many hosts at once is usually higher leverage than fixing isolated single-host issues first.

12.4 Filtering and Searching

Every list page includes a filter bar. Common filters include a minimum severity threshold, a specific host address, or a free-text search matched against vulnerability names.

Note: The default filter on most pages excludes overrides and applies a minimum Quality of Detection (QoD) threshold, commonly 70%. This hides low-confidence detections by default; lowering the QoD filter surfaces more results but with a higher false-positive rate.

13. Vulnerability Analysis

Severity in OpenVAS follows the standard CVSS scale: 0.1–3.9 is Low, 4.0–6.9 is Medium, 7.0–8.9 is High, and 9.0–10.0 is Critical. A result of N/A or Log means the finding carries informational value but was not assigned a CVSS score — these should still be reviewed but do not require the same urgency as scored findings.

13.1 Quality of Detection (QoD)

QoD expresses how confident the scanner is that a given finding is a true positive, expressed as a percentage. A higher QoD means higher confidence; findings below the applied QoD threshold are hidden from default views to reduce false-positive noise.

13.2 EPSS

The Exploit Prediction Scoring System (EPSS) estimates the probability that a given vulnerability will be exploited in the wild within a defined window. Where available alongside CVSS, EPSS helps distinguish a high-severity-but-rarely-exploited finding from a moderate-severity finding that is being actively exploited at scale — both are useful inputs into prioritization, not substitutes for one another.

13.3 False Positives

Not every reported finding represents a genuine issue. Findings should be cross-checked against QoD and, where feasible, manually verified before being escalated as confirmed vulnerabilities, particularly for anything driving urgent remediation work.

14. Risk Prioritization

Prioritization combines CVSS severity, the number of hosts affected, and detection confidence (QoD) rather than relying on severity alone.

14.1 Critical

CVSS 9.0–10.0. Findings in this band typically represent remotely exploitable issues with severe impact (for example, default-credential access to a service). These warrant same-day triage regardless of when they surface in the reporting cycle.

14.2 High

CVSS 7.0–8.9. Serious issues that should be scheduled for prompt remediation, generally within the current reporting cycle, especially where multiple hosts share the same finding.

14.3 Medium

CVSS 4.0–6.9. Should be tracked and remediated on a normal cadence; frequently represents configuration hardening opportunities (deprecated protocol versions, disclosure issues) rather than immediately exploitable flaws.

14.4 Low

CVSS 0.1–3.9. Generally informational or low-impact; addressed opportunistically or as part of broader hardening work rather than urgent remediation.

14.5 Prioritizing by Blast Radius

A vulnerability affecting many hosts is usually a higher-leverage fix than an isolated single-host issue, even when their individual severities are similar. The Vulnerabilities page (Section 12.3) is the right tool for this comparison, since it groups by finding type and shows the affected-host count directly.

15. Host Analysis

Assets → Hosts lists every host that has ever been scanned, along with its current severity rating, detected operating system, and the date it was last modified (rescanned). This view functions as the asset inventory for the scanning program — confirming every expected host across every segment actually appears here and has not silently dropped out of scan coverage.

- Review hosts with a Critical or High severity badge first.
- Review the Modified date for any host that has not been rescanned recently — a stale Modified date on an otherwise-active host can indicate it has fallen out of a target's range.

16. Operating System Detection

Assets → Operating Systems aggregates hosts by detected operating system (via CPE identifiers) and shows severity statistics — latest, highest, and average — per OS, along with how many hosts run each one.

This view is used to spot whether a particular OS family is systematically the source of the worst findings across multiple segments. An OS entry with a Critical Highest score deserves investigation even when its Average severity looks moderate, since a single severe host can be masked by several lower-severity ones in the same OS family.

17. Vulnerability Examples

The following findings illustrate common categories encountered during routine scanning, what each one means in practice, and the standard remediation approach.

17.1 SMB Brute Force Logins with Default Credentials

Severity band: Critical

What it means: The scanner was able to authenticate to an SMB service using default or trivially guessable credentials.

Why it matters: Default-credential access is one of the most severe finding categories, since it typically grants immediate, low-effort administrative or file-share access to an attacker with no exploit development required.

Recommended remediation: Change default credentials immediately, enforce a strong password policy, and disable any accounts that do not require SMB access.

17.2 Microsoft Windows SMB/NETBIOS NULL Session Authentication Bypass

Severity band: High

What it means: The host permits a NULL (unauthenticated) SMB session, which can be used to enumerate users, shares, and system information without valid credentials.

Why it matters: NULL session access is frequently used as a reconnaissance step ahead of more targeted attacks, since it can reveal usernames and share layouts that would otherwise require authentication to view.

Recommended remediation: Disable NULL session access via local security policy or Group Policy, and restrict SMB exposure to only the hosts and segments that genuinely require it.

17.3 FTP Unencrypted Cleartext Login

Severity band: Medium

What it means: The host runs an FTP service that accepts login credentials sent in plain text over the network, with no TLS/SSL encryption wrapping the session.

Why it matters: Anyone able to observe network traffic between the client and this host — on a shared segment, a compromised switch, or via a man-in-the-middle position — can capture the credentials in cleartext and reuse them.

Recommended remediation: Replace plain FTP with SFTP or FTPS. If FTP must be retained for legacy compatibility, restrict access to a trusted management segment and disable it from any segment reachable by general users.

17.4 SSL/TLS Deprecated Protocol Detection (TLS 1.0 / 1.1)

Severity band: Medium

What it means: The host accepts connections using outdated, deprecated TLS versions that contain known cryptographic weaknesses and are excluded from current compliance baselines.

Why it matters: Deprecated TLS versions are vulnerable to several published downgrade and cryptographic attacks and signal that a service's TLS configuration has not been reviewed recently.

Recommended remediation: Disable TLS 1.0 and 1.1 and enforce TLS 1.2 or higher exclusively, after confirming client compatibility.

17.5 DCE/RPC and MSRPC Services Enumeration

Severity band: Medium

What it means: The host exposes the DCE/RPC endpoint mapper service, typically on port 135/tcp, which can be queried remotely to enumerate available RPC services.

Why it matters: While not a vulnerability by itself, RPC enumeration is a common early reconnaissance step that reveals which services and interfaces are running on a host.

Recommended remediation: Restrict access to port 135/tcp and related dynamic RPC ports to trusted internal management hosts only; do not expose DCE/RPC endpoints to general user segments.

17.6 ASP/PHP Source Code Disclosure (Active Check)

Severity band: Medium

What it means: A web server misconfiguration, commonly involving how special or escaped characters in a URL are handled, causes the server to return raw, unprocessed script source instead of executing it.

Why it matters: Source code can reveal internal logic, hardcoded credentials or connection strings, and implementation detail that an attacker can use to plan further attacks.

Recommended remediation: Apply the relevant vendor patch for the web platform and ensure URL-encoding edge cases are handled by an up-to-date request parser.

17.7 TCP Timestamps Information Disclosure

Severity band: Low

What it means: The host responds to TCP probes with timestamp values that can be used to estimate system uptime and assist OS fingerprinting.

Why it matters: On its own this is a low-severity informational disclosure rather than an exploitable vulnerability; it becomes more relevant as one data point among several during broader reconnaissance.

Recommended remediation: Generally an accepted, low-priority finding. Where fingerprinting-resistance is a hard requirement, TCP timestamps can be disabled at the OS network-stack level.

18. Security Recommendations

- Prioritize remediation by combining severity, affected-host count, and QoD confidence rather than severity alone.
- Treat Critical and High findings as same-cycle work items rather than batching them into a periodic summary.

- Enable authenticated scanning wherever credentials can be safely provisioned; unauthenticated scans systematically under-detect local vulnerabilities such as missing OS patches.
- Disable legacy protocols (unencrypted FTP/Telnet, deprecated TLS versions) proactively rather than only in response to a finding.
- Restrict administrative and RPC-style services to dedicated management segments rather than general user network segments.
- Re-validate target IP ranges whenever segment infrastructure changes; stale ranges are the most common cause of incomplete scan coverage.
- Treat exported reports as sensitive documents and distribute them only through approved, access-controlled channels.

19. Troubleshooting

19.1 No Hosts Detected

Likely causes: an incorrect or outdated IP range in the target definition; hosts offline at scan time; or a firewall between the scanner and the target range blocking host-discovery probes.

Resolution: confirm the target range against the current asset list, confirm hosts are powered on and connected, and verify firewall rules permit scanner-to-target traffic in both directions. Re-run a quick Host Discovery scan to isolate a targeting issue from a connectivity issue.

19.2 Severity Shown as N/A

Likely causes: a discovery-type scan configuration was used, which does not perform the deeper checks needed to produce CVSS-scored findings; or the scan completed with only Log-level results.

Resolution: this is expected behavior for discovery-type profiles, not an error. If N/A appears on a Full and Fast scan where scored findings were expected, check the Results tab directly rather than relying on the summary badge.

19.3 Scan Returns Zero Findings

Likely causes: hosts were unreachable during the scan window; an overly narrow port list was attached to the target; authentication credentials for authenticated checks were missing or incorrect; or the scan was stopped or interrupted before completion.

Resolution: confirm task status was Done rather than Stopped or Interrupted, confirm a comprehensive port list is attached, and verify credentials if authenticated checks were expected.

19.4 Scanner Unavailable

Likely causes: the scanner service is not running, is overloaded with concurrent tasks, or has lost connectivity to the management server.

Resolution: check Configuration → Scanners for status, and reduce the number of simultaneously running tasks if the scanner appears overloaded.

19.5 Feed Synchronization Issues

Likely causes: the NVT, CERT, or SCAP feed has not updated recently.

Resolution: check Administration → Feed Status for the last successful sync, and confirm connectivity from the scanner host to the feed server if it is significantly out of date.

19.6 Slow Scans

Likely causes: a large host count combined with a comprehensive profile's test set; multiple tasks competing for scanner resources; or unresponsive hosts causing repeated timeouts.

Resolution: stagger large scans rather than running them simultaneously, use a narrower profile where full depth is not required, and investigate consistently unresponsive hosts.

20. Best Practices

- Keep target names aligned with segment names so tasks, reports, and dashboards stay readable at a glance.
- Default to the Full and Fast profile for routine scans; reserve narrower profiles for specific, time-sensitive needs.
- Use the Vulnerabilities page, not just the Reports page, when prioritizing remediation — affected-host count is as important as raw severity.
- Export and archive a copy of every periodic report, even when no Critical/High findings are present, to maintain a complete historical record.
- Check feed synchronization status before any major scanning cycle.
- Stagger concurrent scans across large target groups to avoid scanner overload.
- Keep environment reference tables in sync with the live Targets configuration — update both together, never one without the other.

21. Conclusion

A recurring OpenVAS deployment, organized around clearly scoped targets and a disciplined review cadence, turns vulnerability scanning from a one-off exercise into a sustained risk-reduction program. The workflow described in this guide — environment preparation, target and scan configuration, execution, monitoring, and risk-based prioritization of results — generalizes across environments of varying size and technology mix, and provides a foundation that can be extended with authenticated scanning, scheduled automation, and integration with broader security operations tooling as a program matures.